

Lesson 4: How does the speed (and the mass) of the vehicles in a collision affect the outcome?

Navigate

Even when you cannot prevent a collision, braking can help slow your car down before it collides with another car.

With your class



1. How will the speed of a vehicle at the moment of first contact in a collision affect the outcomes?

Analyze Data

With your class



2. Make some initial observations of what happens when two carts collide together in three different collisions.

Construct Initial Explanations

Turn and talk



3. What is happening during these collisions that is causing the yellow-green cart (left) to slow down and the blue cart (right) to speed up?
 - Be ready to share your ideas with the whole class.

Make Predictions

With your class



4. Take a poll: If we could measure the contact force(s) between the carts during the collision how would we expect they would compare?
 - Only cart A would be pushed on by the other cart.
 - Only cart B would be pushed on by the other cart
 - Both carts A and B would be pushed on, but the strength of the forces on each would be different.
 - Both carts A and B would be pushed on, and the strength of the forces on each would be the same.

Analyze and Interpret Data

With your class



5. Watch the short video to orient to the how sensors these carts collect force and velocity data before, during, and after a collision.

With a partner



6. Annotate your handout to help identify any patterns you notice across the three collisions.
- Be ready to share with the whole class.

With your class



7. Summarize the patterns you noticed across the three collisions in this part of the data. Reference slide.

Interpreting “Negative Force”

Turn and talk



8. What does a negative force mean?

Analyze and interpret data

With your class



9. Summarize the patterns you noticed across the three collisions in this part of the data. Reference slide.

Interpreting “Negative Speed”

Turn and talk



12. What does a negative speed mean?

Recording a Definition of Velocity

In your notebook



13. Add entries to your personal glossaries to reflect these ideas.
- How is velocity different from speed?
 - How have we expanded on our understanding of force since the last unit?

Use Mathematical Thinking

On your own



14. Record these on your exit ticket as well as on the bottom of your handout and store your handout in your notebook.
- Which collision produced the largest velocity change (ΔV) for each cart?
 - How much did the velocity of each cart change in that collision?

With your class



15. Which collision produced the largest velocity change (ΔV) for each cart?

16. How much did the velocity of each cart change in that collision?

Analyze data

With your class



17. Rank order the 3 collision them from where the velocity changes were the largest to the smallest

18. How do the magnitude of the forces on the carts compare for this ranking?

Develop Initial Explanations

In each collision:

- The magnitude (size) of the forces on both carts were equal to each other across every point in time.
- The magnitude (size) of the velocity changes for both carts were equal.

With your class



19. What is happening during every collision that is causing this symmetry in outcomes?

Make Predictions

On your own



20. Record your predictions for the 3 collision listed below on a new sheet for your notebook. We will keep the initial velocities of the carts the same as in Collision A and change only the mass of the carts.

- Collision D: we will double the mass of both carts.
- Collision E: we will double the mass of the green (left) cart but not the blue (right) cart.
- Collision F: we will double the mass of the green (left) cart but not the blue (right) cart.

Analyze Data

With a partner



21. Watch the video of collisions D, E, & F.

22. Analyze the patterns in the related sensor data on the related handout and compare these to collision A.

23. Annotate the handout to help you figure out how changes in mass affected the outcomes of the collision.

Argue from Evidence

With your class



24. How does mass affect the forces on each object in a collision?
25. How does mass affect the change in velocity of each object in a collision?

Looking for a Relationship

With your class



26. Reference collision E and collision F graphs.

With your class



27. For collision E graph: $v_{Gi} = .79 \text{ m/s}$, $v_{Gf} = .33 \text{ m/s}$, $v_{Bi} = 0 \text{ m/s}$, $v_{Bf} = .92 \text{ m/s}$
- Reference slide
28. For collision F graph: $v_{Gi} = .8 \text{ m/s}$, $v_{Gf} = -.12 \text{ m/s}$, $v_{Bi} = 0 \text{ m/s}$, $v_{Bf} = .46 \text{ m/s}$
- Reference slide

Use Mathematical Thinking

Turn and talk



29. Based on the patterns we have figured out so far, what do you predict would happen if the mass of one cart was five times the mass of the other cart and they collided?

Navigate

With your class



30. What does this relationship predict the outcome would be for a more extreme mass ratio collision?

Navigate

With your class



31. Take a poll: is there a similar mathematical relationship between the masses and the changes in velocity for two objects in any of these other collision conditions?